Class 3: Bed Configurations (Part 1)



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Emeritus MIT

What a strange, strange boy

He sees the cars as sets of waves
Sequences of mass and space

Joni Mitchell,

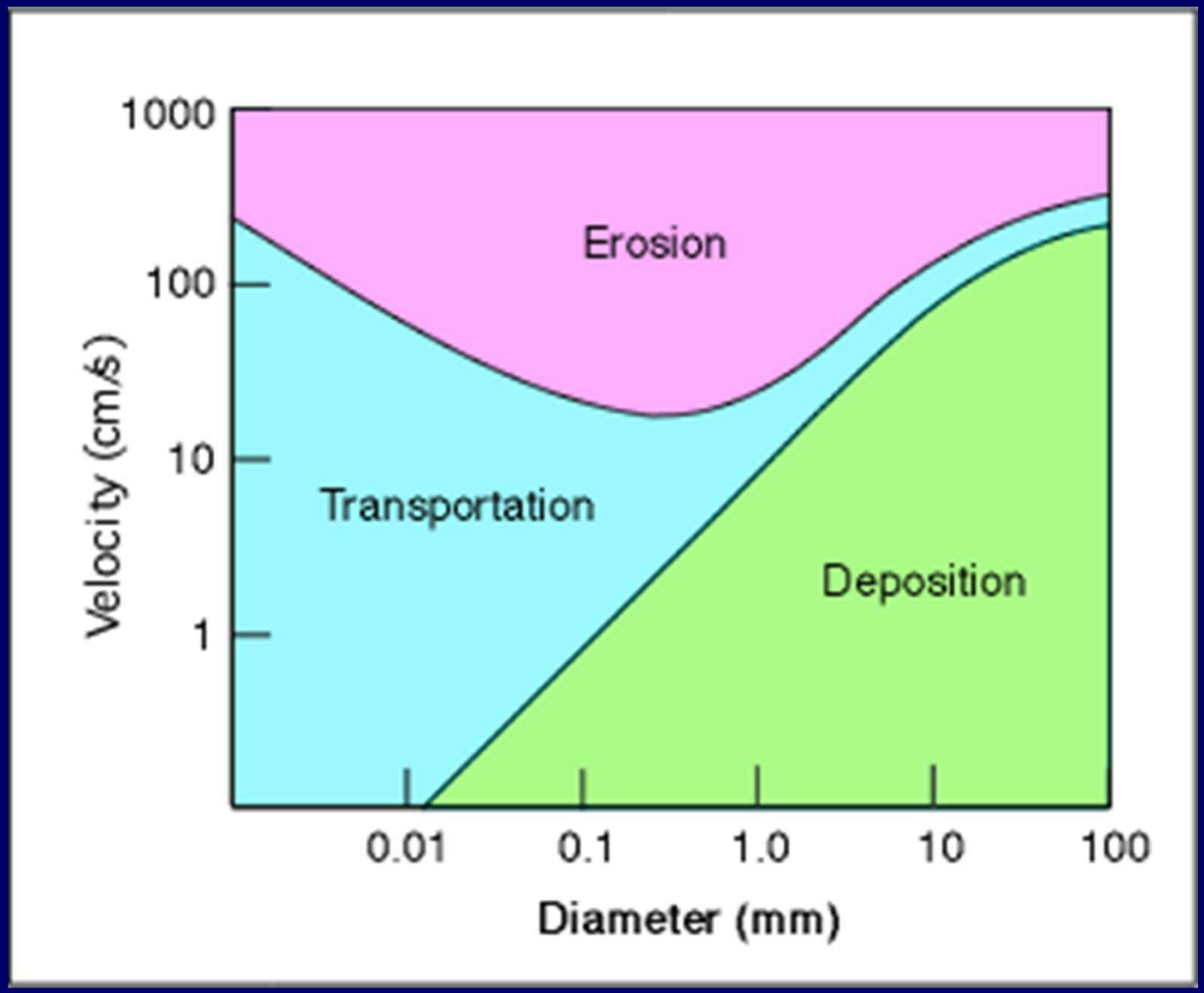
'A strange Boy'

Hejira, Asylum Records

Class outline: How do water flows interact with a bed made of sand?

- What drives erosion and deposition?
- Describe the bedforms that develop in fine-coarse sands under unidirectional flows

What controls erosion and deposition of a sediment bed?



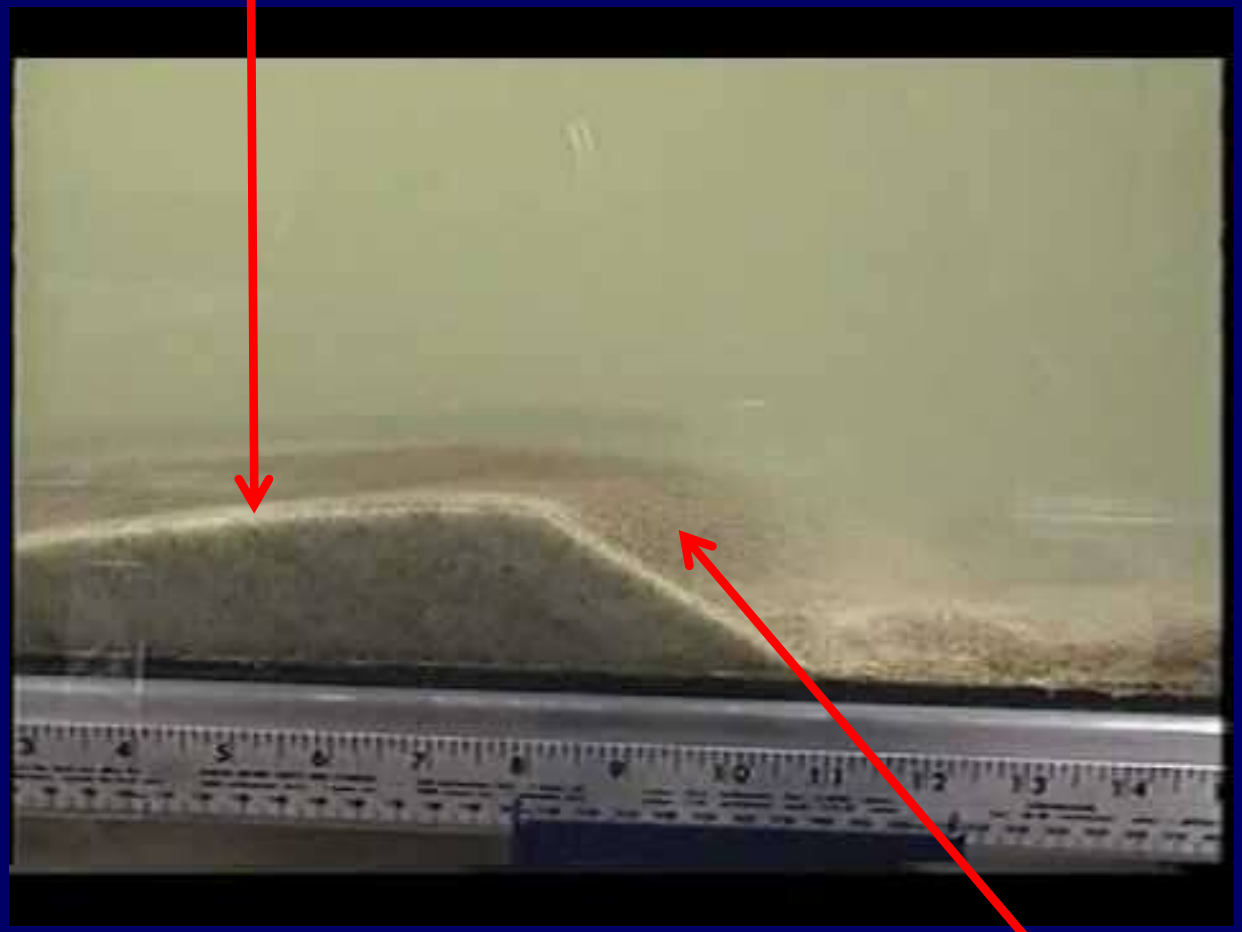
One viewpoint:
Erosion and deposition is controlled by the velocity of the flow

Implication of diagram:
High velocity flows are erosional and low velocity flows are depositional.

We know by studying sedimentary structure in the flume and stratigraphic record this is not true.

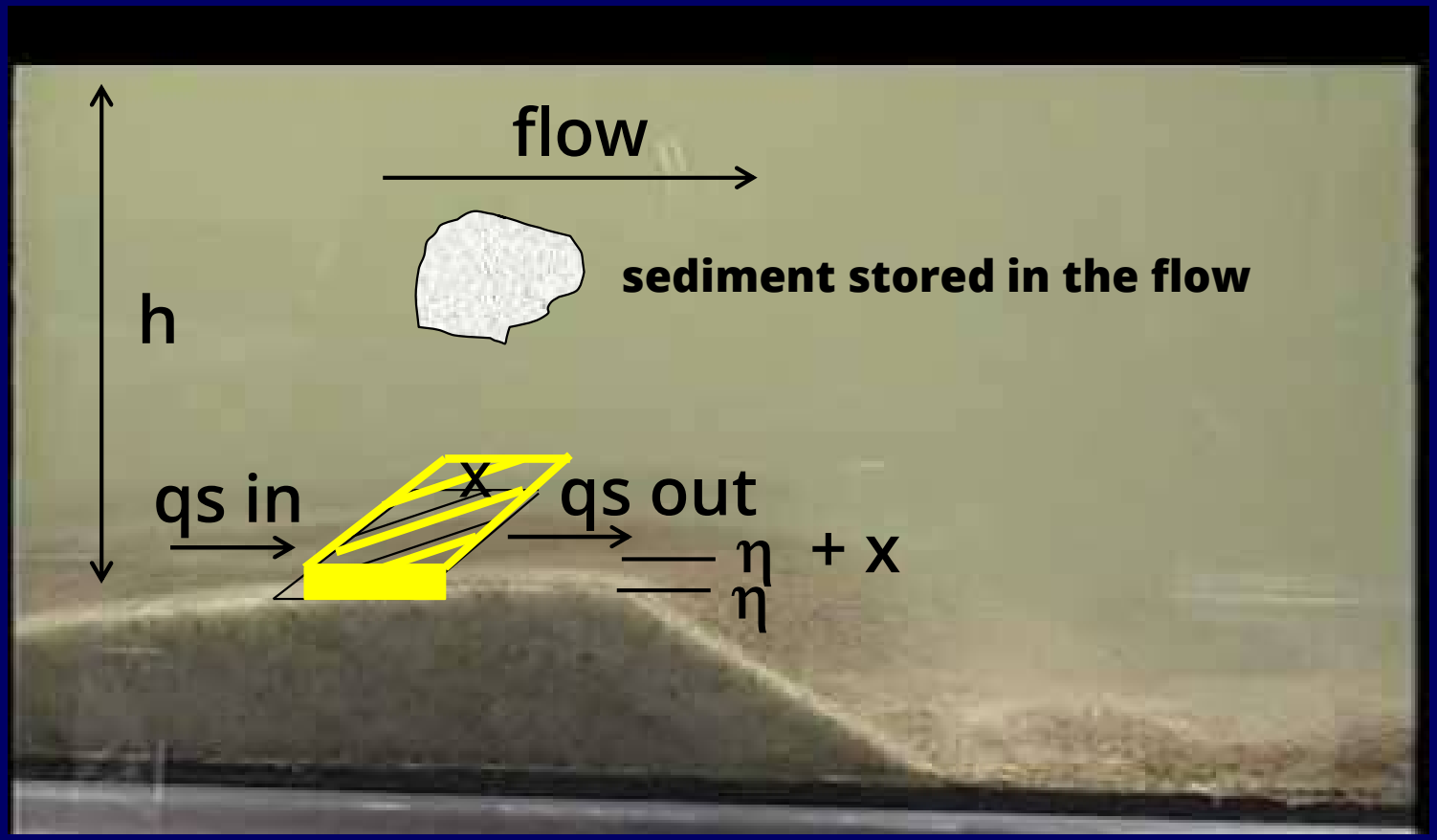
Sediment mass balance: Exner equation

goal: relate dynamics of topography.....



.... to sediment flow over the bed

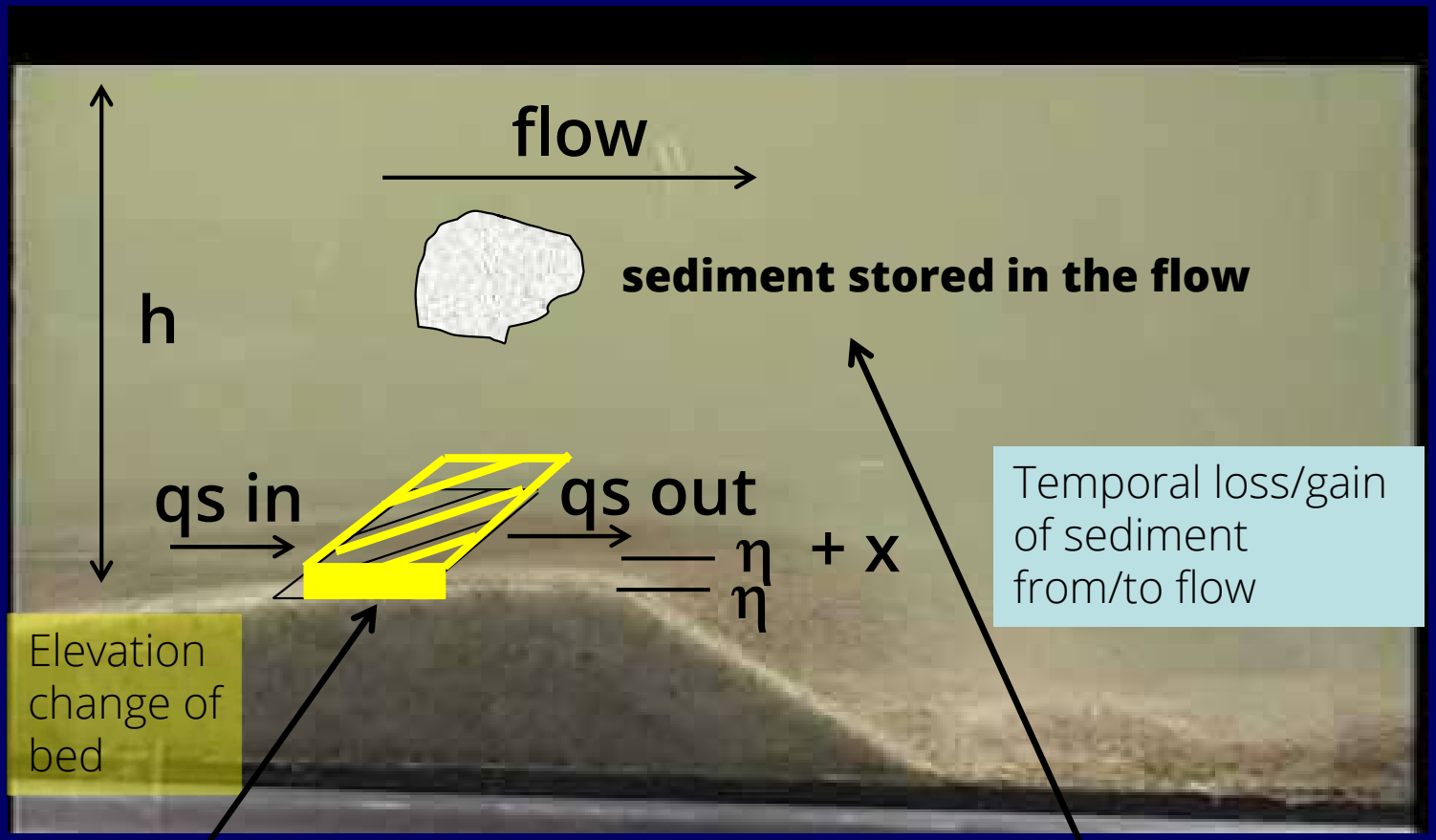
Alternative viewpoint: Sediment mass balance



qs: volume sediment flux per unit width (x) per unit time
h: flow depth
η: bed elevation

$$\partial q_s = q_s\text{out} - q_s in$$

Sediment mass balance: Exner equation



qs: volume sediment flux per unit width per unit time
h: flow depth
η: bed elevation

$$\partial q_s = q_s\text{out} - q_s in$$

$$c_o \frac{\partial \eta}{\partial t} + \frac{\partial q_s}{\partial x} + \frac{\partial}{\partial t} \int_{\eta}^{\eta + h} c(y)\, dy = 0$$

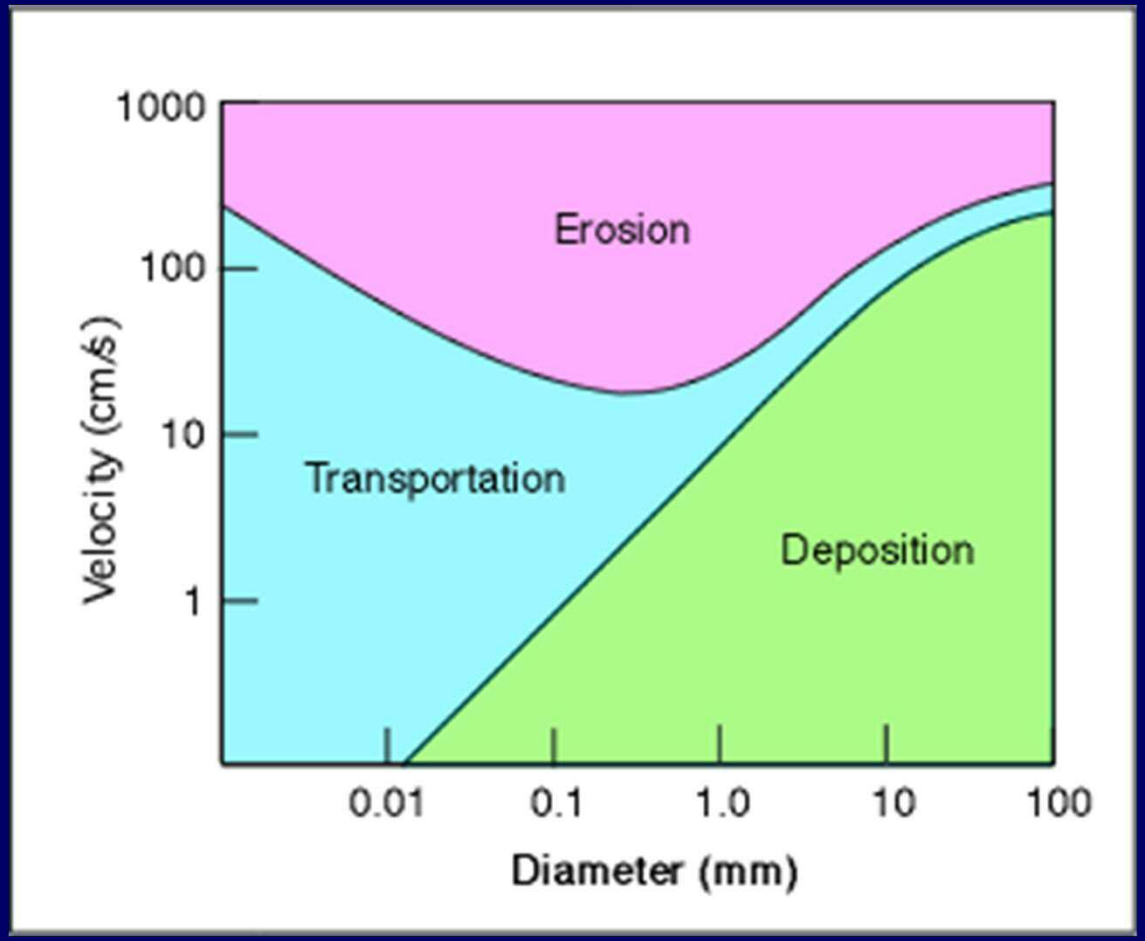
Spatial change in sediment flux

total sediment in flow

Implication of sediment mass balance: Exner equation

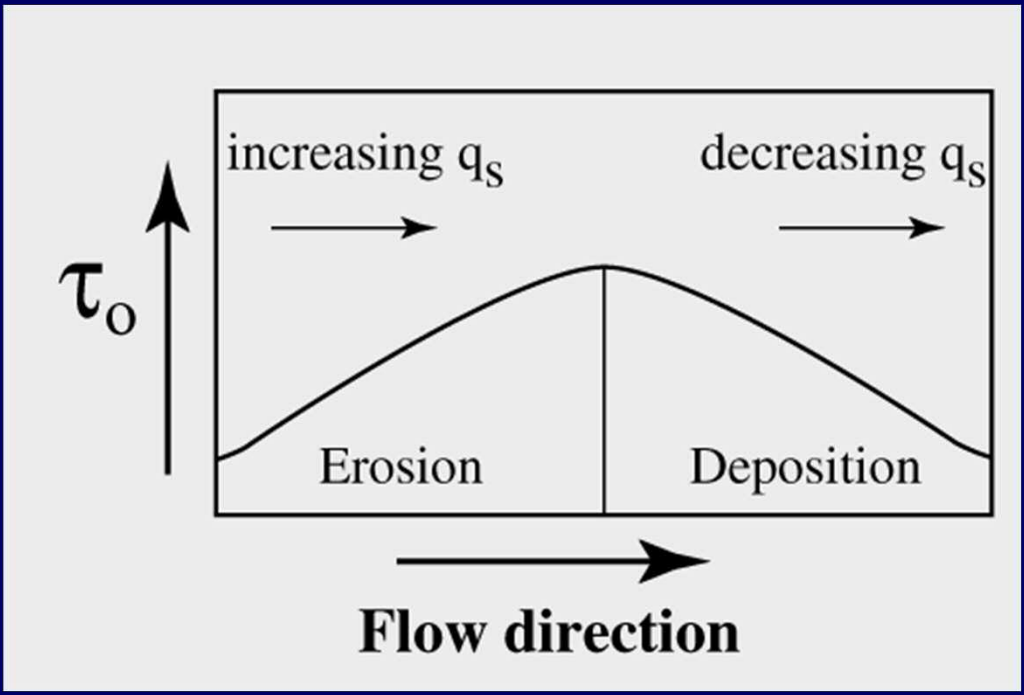
- Threshold concept: Erosion vs deposition is controlled by the rate of change (in space and time) of sediment load

- In particular, it is NOT true that fast flows are erosional and slow flows are depositional (e.g., as implied in the common application of the Hjulstrom diagram)



Significance of turbulence for erosion and deposition

- Turbulence results in variation in bed shear stress along and across flows
- Forces can vary by factors of four or even more with time
- Time scales for change in forces on a sediment particle can be very short, fractions of a second
- Bed shear stress governs the power of the current to move sediment; erosion and deposition depend on the change in boundary shear stress in the downstream direction
- In general, sediment transport rate (q_s; the amount of sediment that is moved by a current) increases with increasing bed shear stress

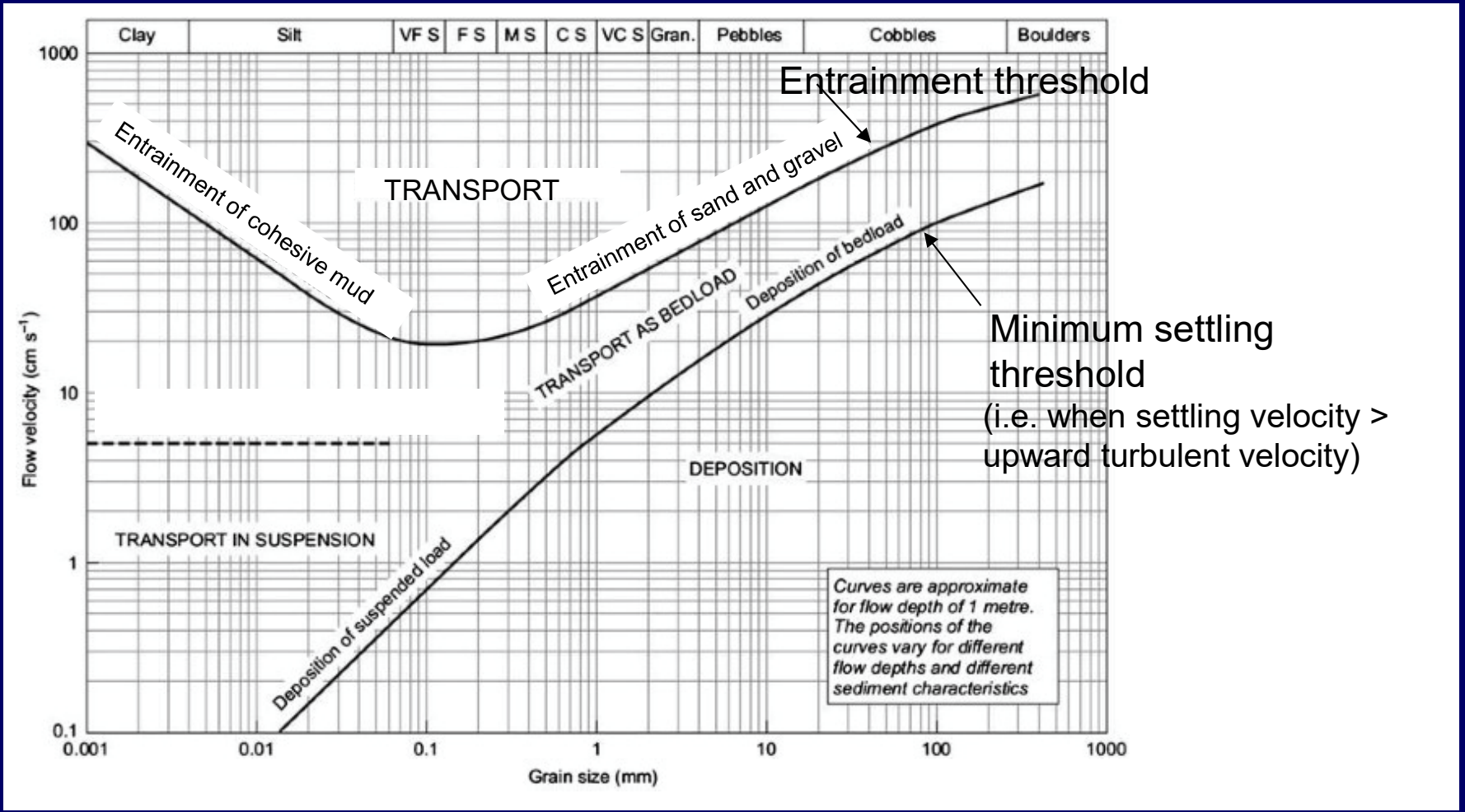


Variation in τ_o along and across the flow due to turbulence leads to a pattern of erosion and deposition on the bed of a mobile sediment

Entrainment Thresholds for Movement of Sediment

Improved Hjulstrom diagram

$f(\rho, \mu, d_s, \rho_s, g, \tau_0)$



Concerns the minimum shear stress to get a grain to move in a flow (i.e. τ_{cr})

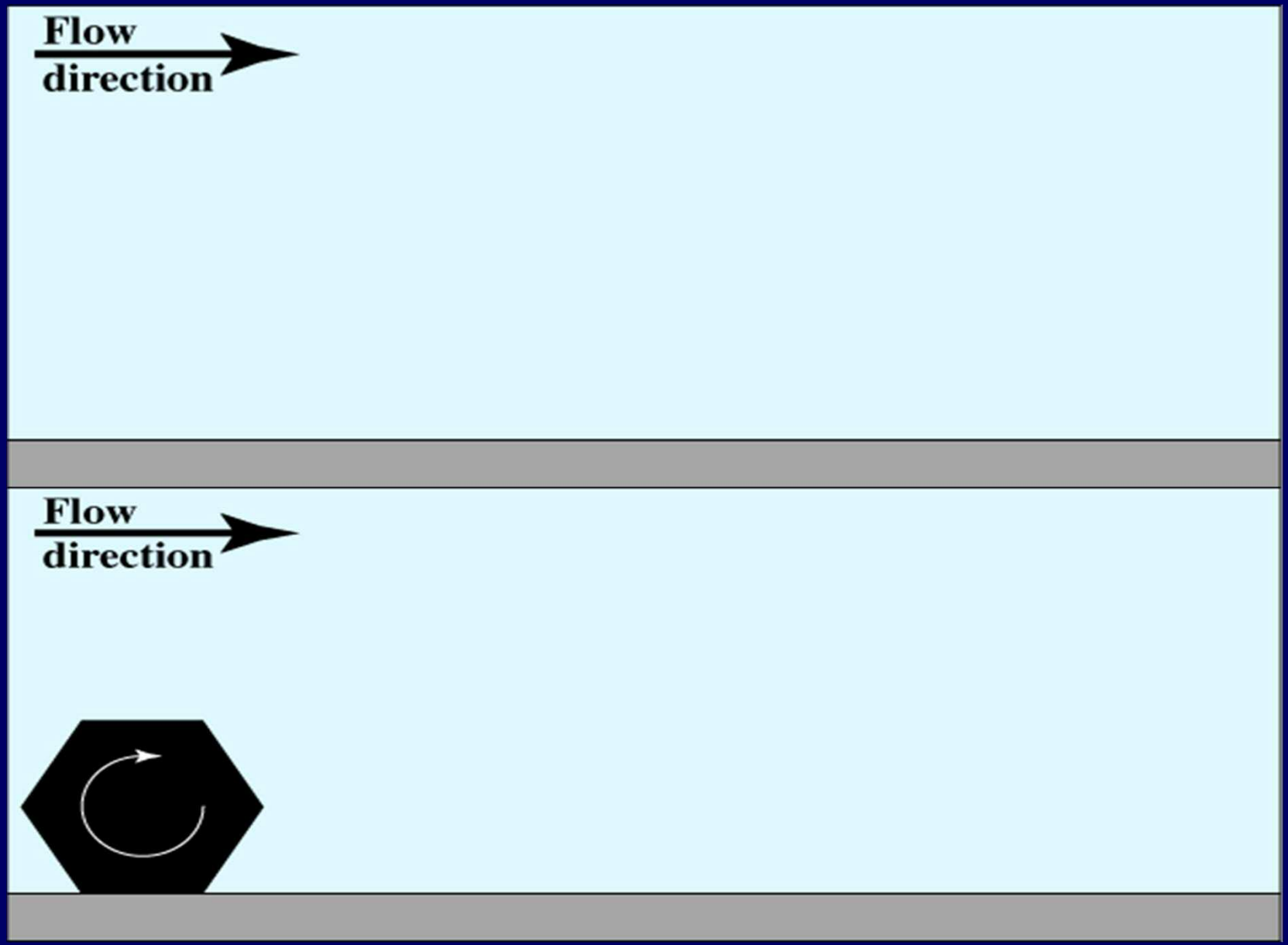
In general, large grains require higher τ_{cr}.

[Flow Competence: faster moving flows can carry *larger* particles

Flow Capacity: faster moving flows can also carry *more* particles]

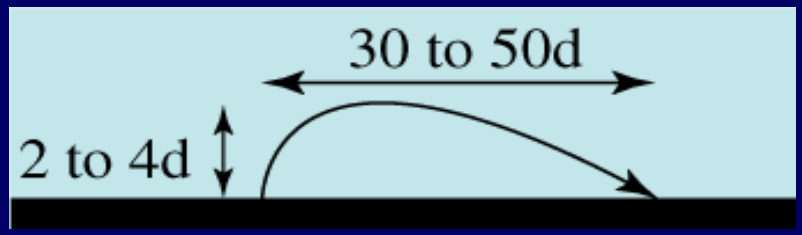
Mode of movement: Contact load

$\tau_0 > \tau_{cr}$



Mode of movement: Saltation

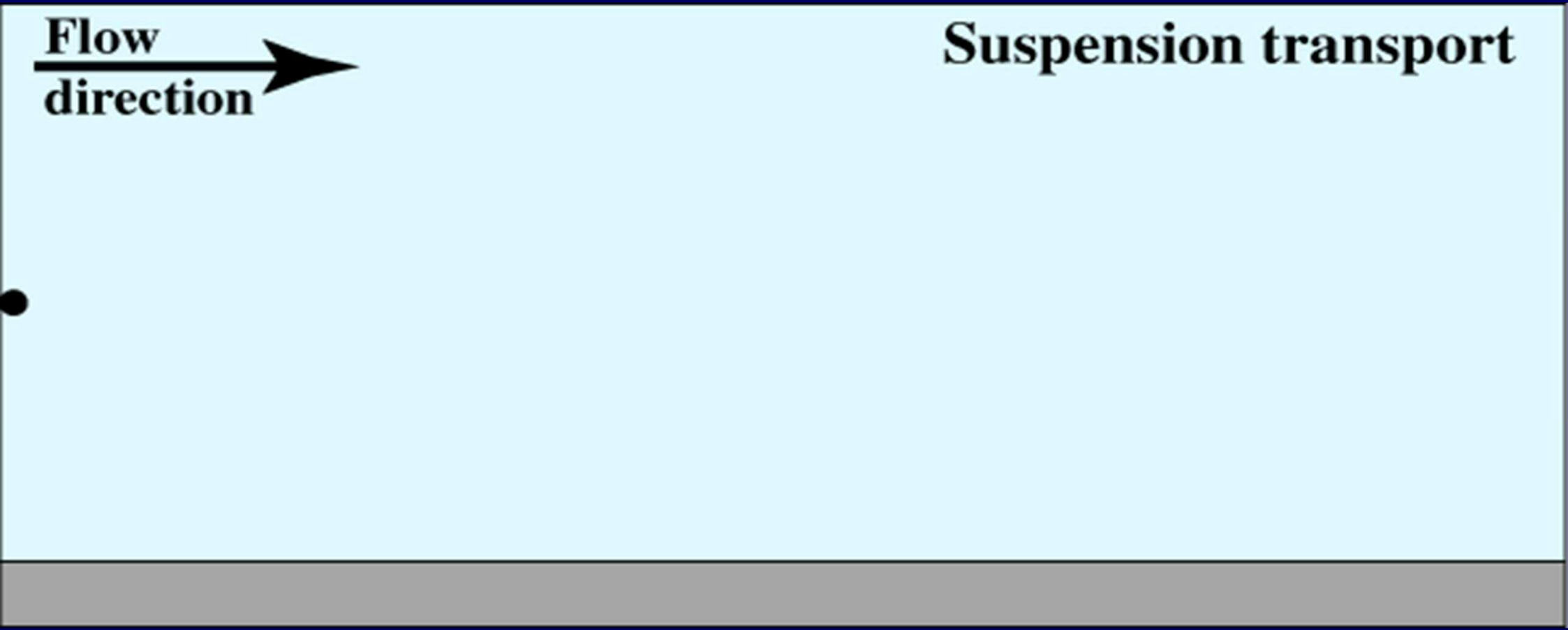
$\tau_0 >> \tau_{cr}$





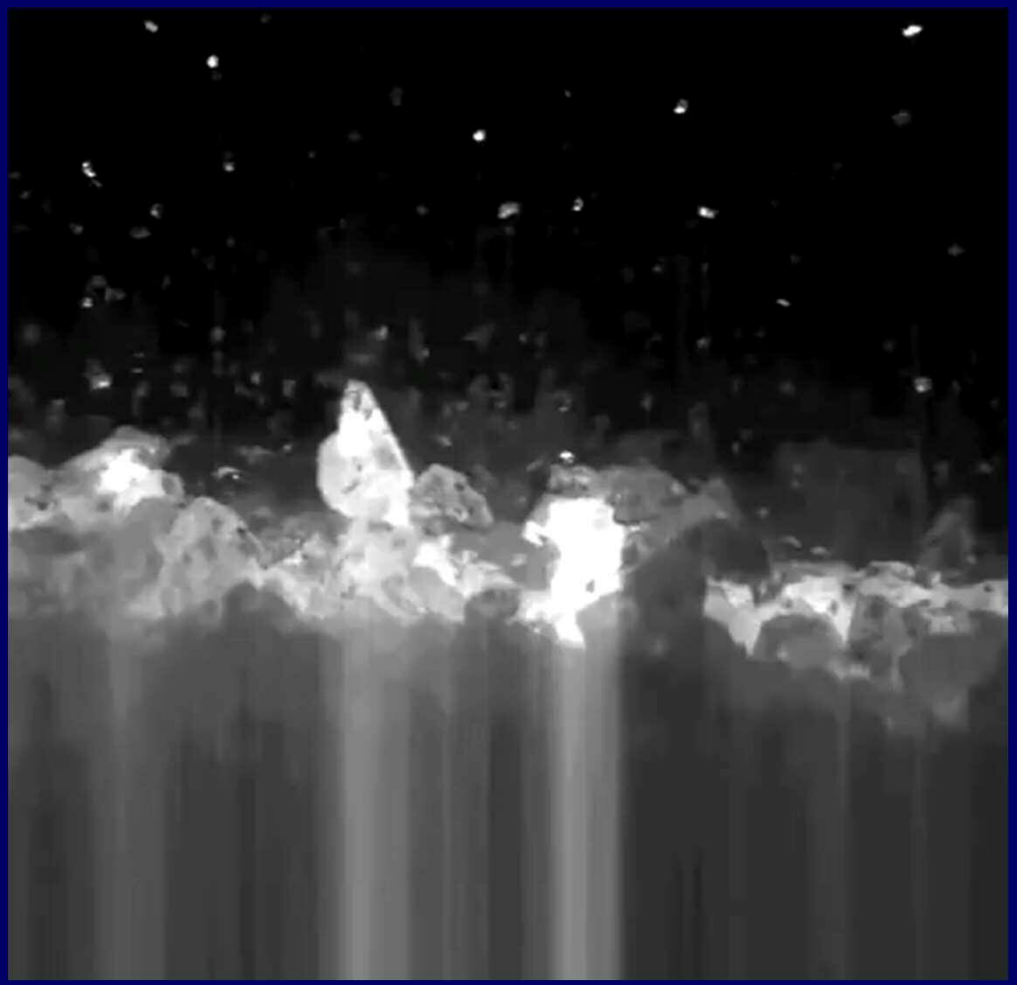
Mode of movement: Intermittent suspension

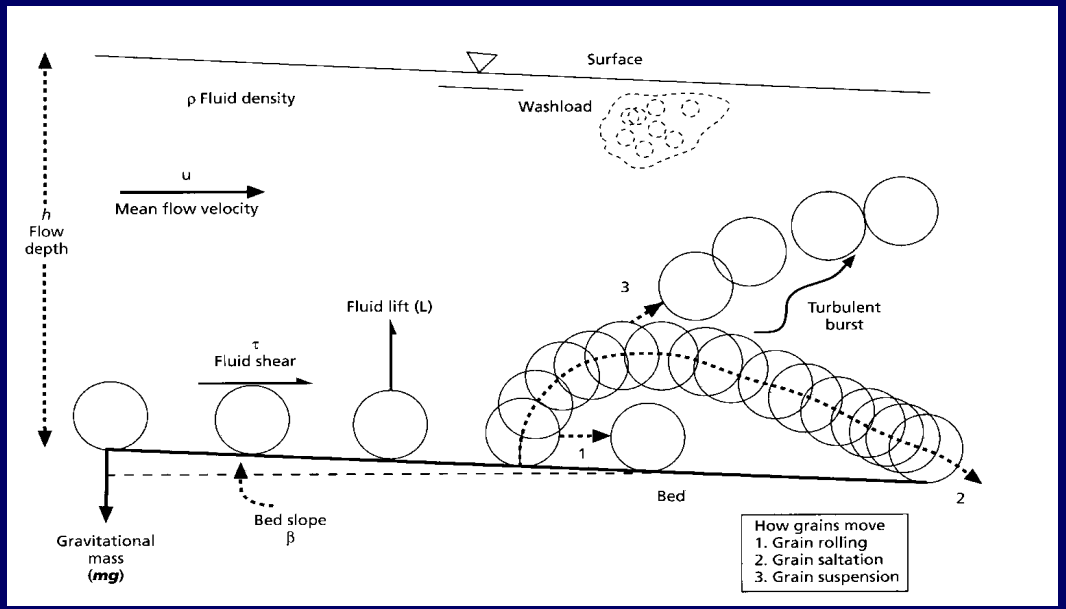
$\tau_0 >>> \tau_{cr}$



Modes of transport in sediment

A flow exerts forces on all of the sediment grains exposed at the surface of a loose sediment bed





In a population of particles of different sizes, the different sizes will move in different ways

Bedforms

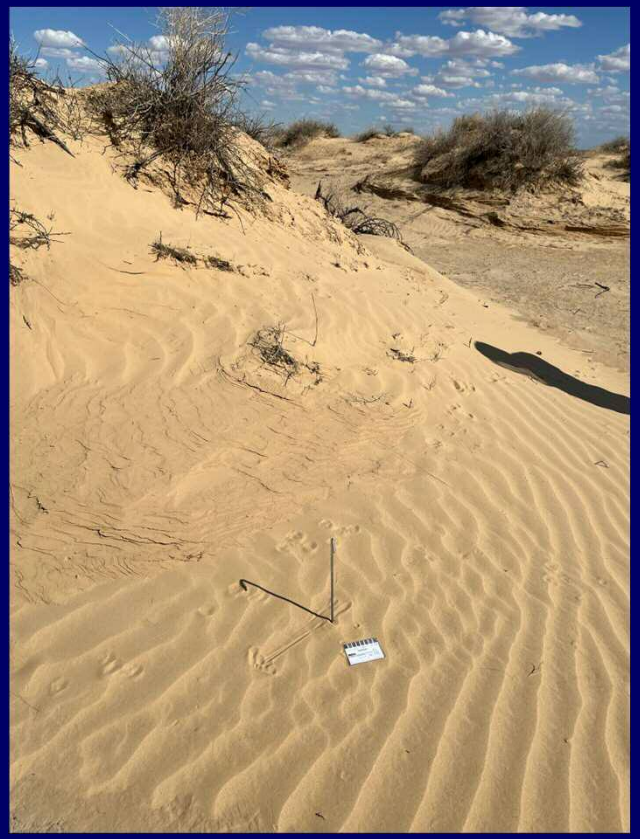
Sediment transported over a bed by a turbulent flow is moulded into bedforms

- orders of magnitude larger than grains
- occur under a wide range of conditions

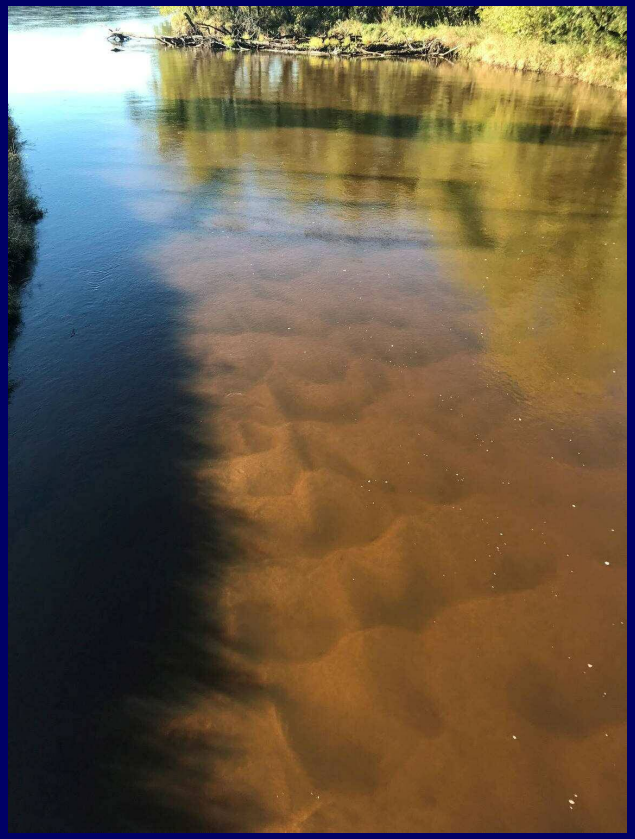


Wave ripples on a shoreline

Examples of bedforms

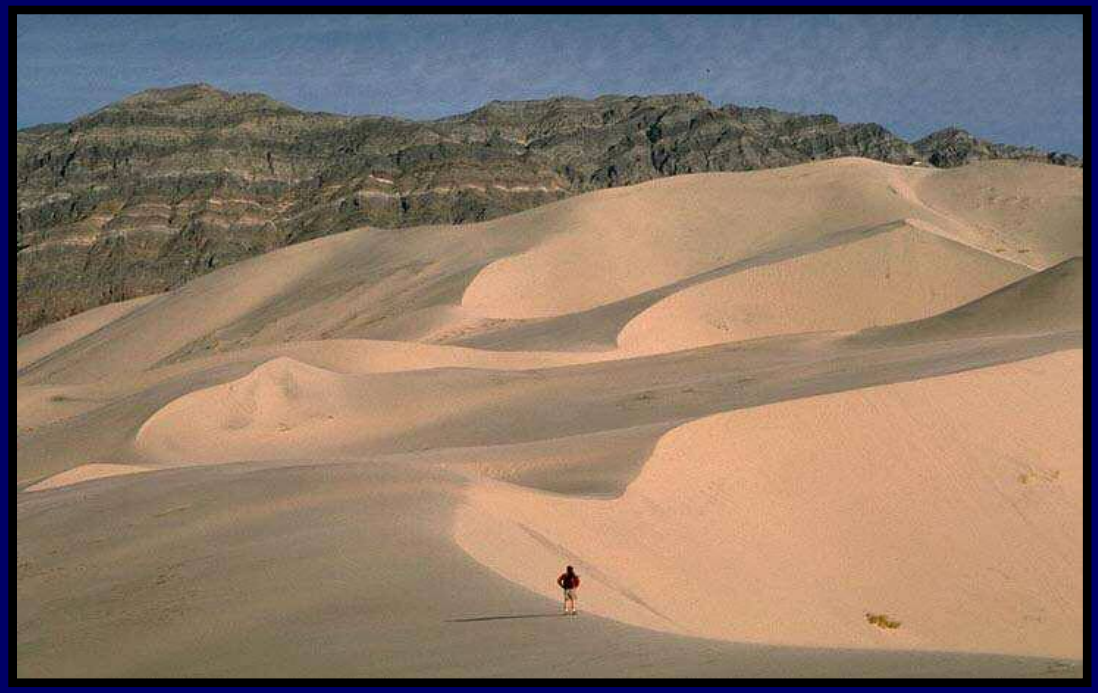


Wind ripples in a desert



Dunes on a river bed

Terminology: Bedforms vs Sedimentary Structures



Bed configuration made of of bedforms (e.g. dunes)

Guide to interpretation:

(1) to observe and describe the structures carefully, then

(2) infer from internal structure type of bedform that produced it, migration pattern, rate of deposition

(3) to interpret the flow conditions responsible for producing the things you inferred in step 2.

Bed configuration: overall bed geometry that exists at a given time in response to the flow, and is composed of

Bedforms: individual topographic elements

Sedimentary structures: record of deposition in the presence of bedforms



Cross-bedding (sedimentary structure) produced by the migration of a dune bedform

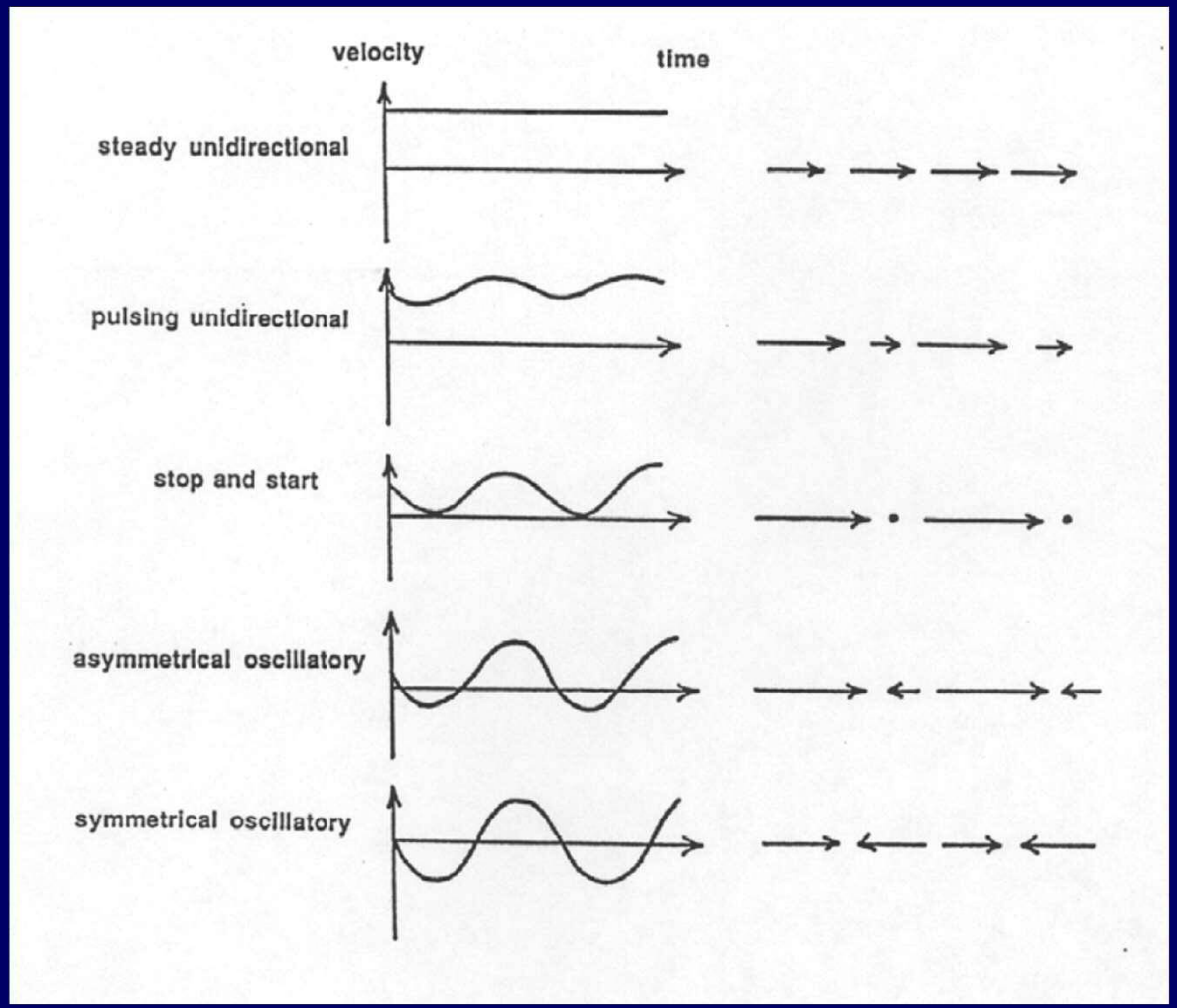
The type of flows that produce bedforms

Definition of a bedform: produced by the accumulation of mobile sediment grains into some shape or morphological element due to local changes in deposition and/or erosion in response to the interaction with a flowing current of air or water

Bedforms interact with flow
- create form drag which slows flow (drag coefficient C_f), sheds turbulence

The flows that make bed configurations can be complex and vary with:
- direction of flow
  - Unidirectional
  - Pure oscillatory
  - Flows with some net translation
  - Combined
- duration of flow
- strength of flow (variation through time)

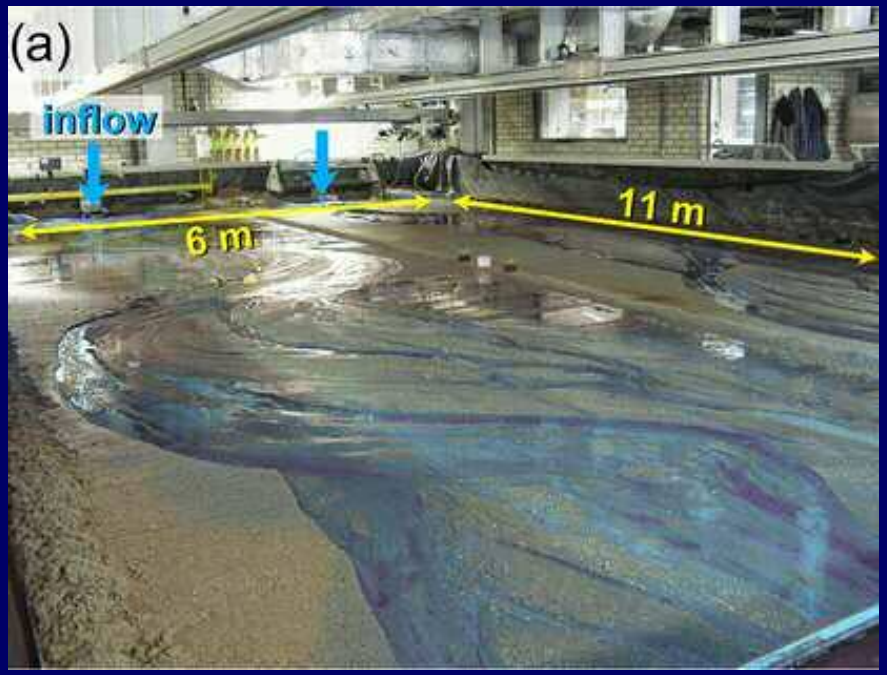
Other important factors
- grain diameter and density of sediment
- the nature of turbulent boundary (hydraulically smooth or rough)



Spectrum of kinds of unidirectional and oscillatory flows, and that could be combined to make a combined flow

Experimental studies of bedforms and bed configurations

Lab experiments on bedforms



Tanks to study bed configurations in unidirectional flows (e.g. rivers)



The Deep- wave tank

If the flow changes with time the bed configuration adjusts in response but in natural flows bed configuration lags behind the change in flow – usually the bed configuration and flow are in disequilibrium

Most bedforms are in sands, but they are also found in gravel and silts

Bedforms are important in both geology and engineering

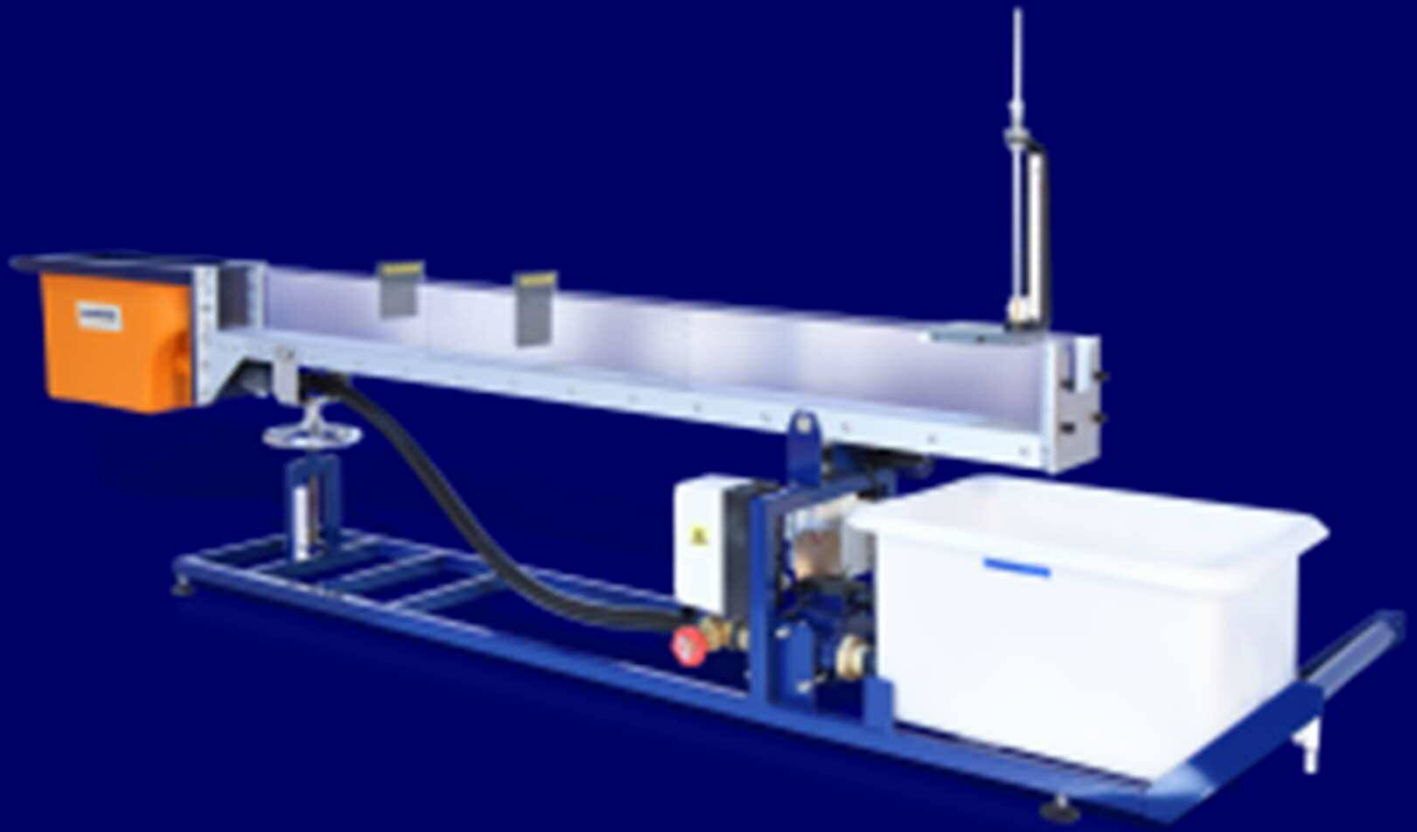
The utility of bedforms and their internal sedimentary structures





- Provide [loose] constraints on flow depth and velocity
- Indicate wave-, wind-, tide-, and unidirectional current-dominated flow conditions
- Palaeocurrent (or palaeoshoreline) indicators
- Way up indicators (important for tectonic studies)

An introduction to the flumes



Controls to adjust

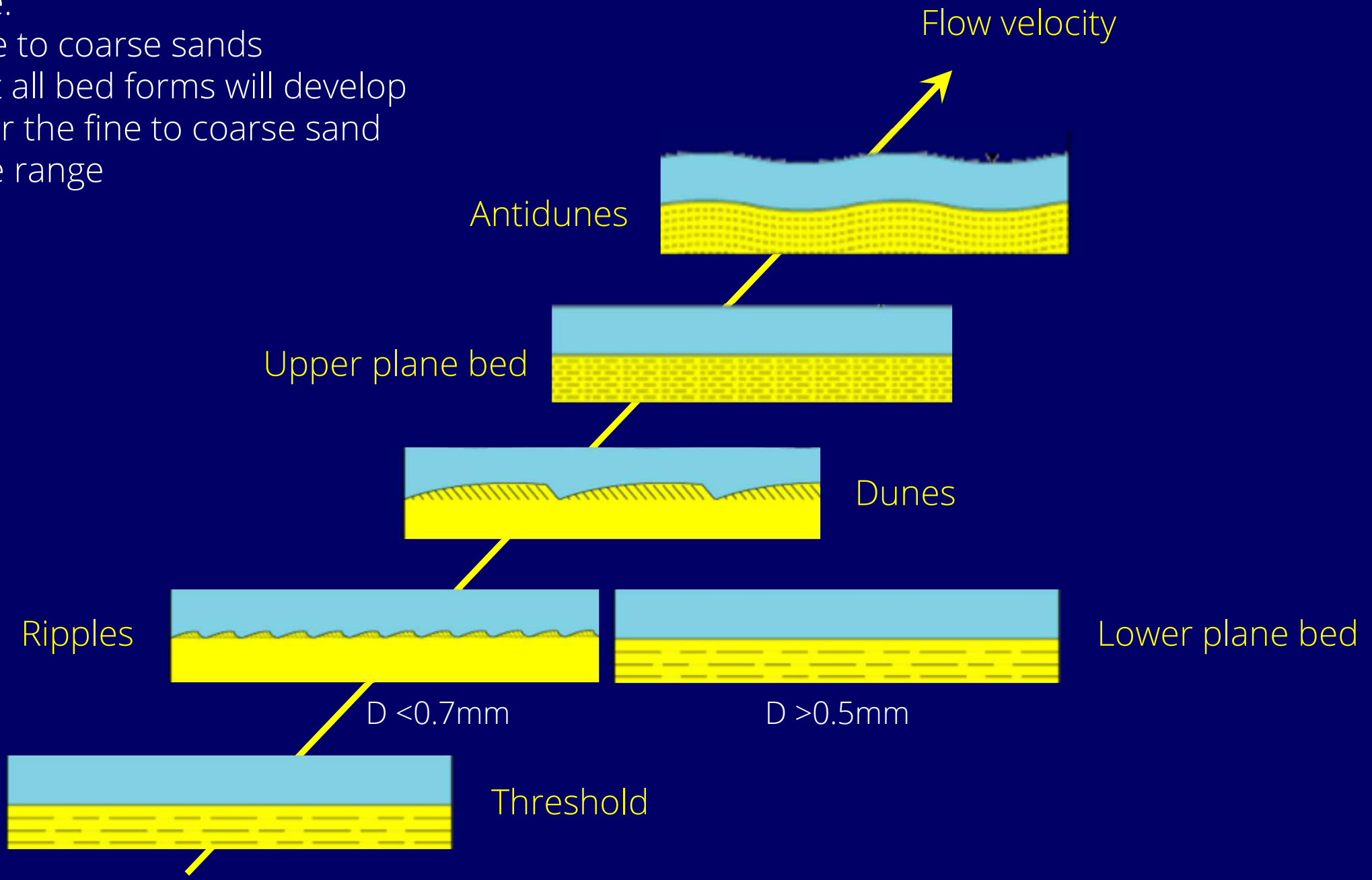
- slope
- discharge (pump and valves)
- sediment characteristics

Techniques to quantify

- slope (water and bed)
- flow velocity (mean, surface and bed)
- flow depth
- sediment transport rate
- sediment characteristics

Sequence of bed forms produced under unidirectional flows

Note:
- fine to coarse sands
- not all bed forms will develop over the fine to coarse sand size range



Lower Plane Bed: a bed with active transport but no obvious topography

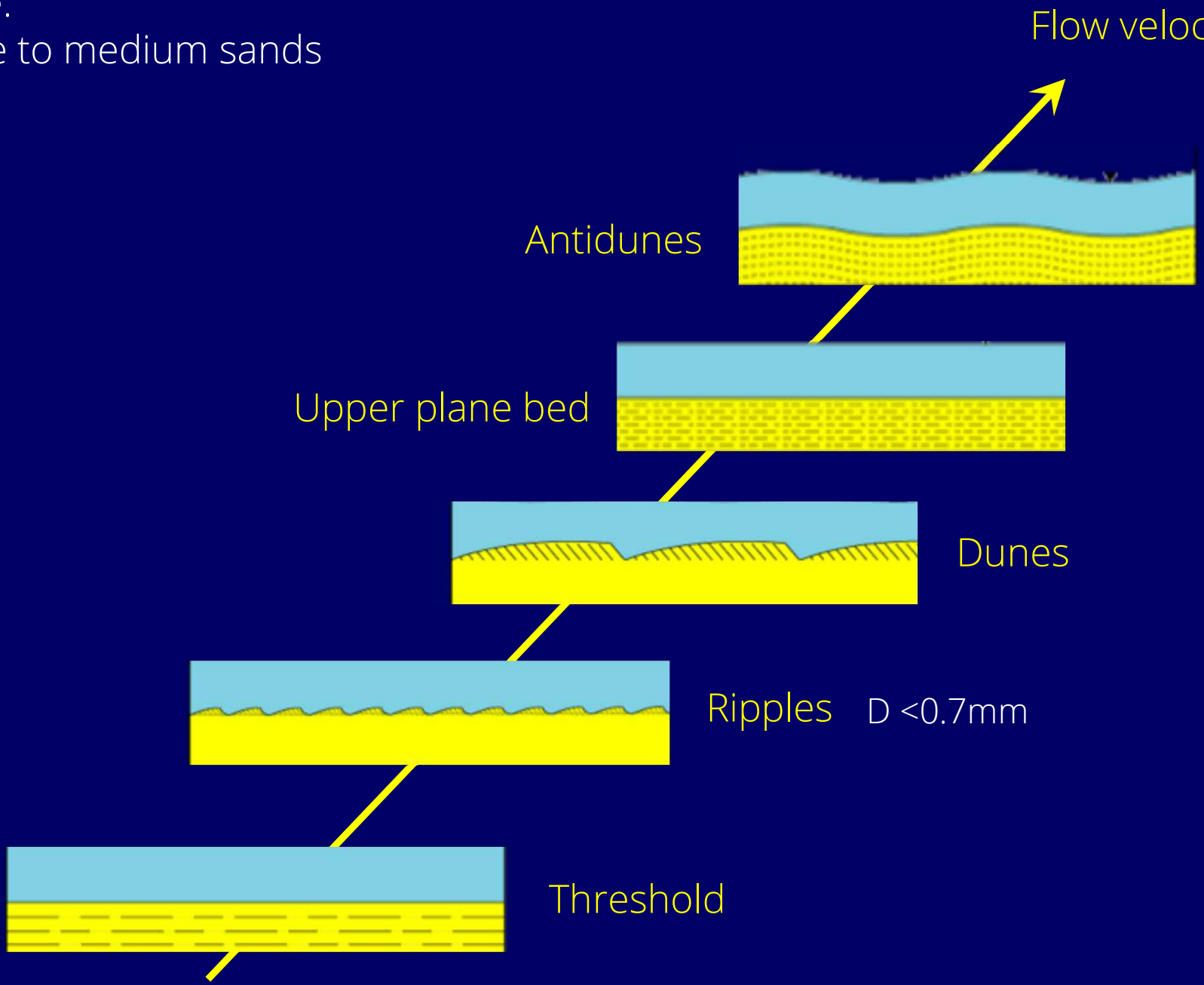


Gravelly streaks mantling top of sandy bar in a river (some dunes also present)
Field of view ~15 x 15 m
(Mohrig, 2006)

- Form at threshold of bed load motion
- Approximately planar surface
- Grains transported by a mixture of rolling and saltation (as contact load)
- Mean grain size must be >0.5 mm
  - Hydraulically rough

Sequence of bed forms produced under unidirectional flows

Note:
- fine to medium sands



Flow structure in the flume at the start of the experiment

The water surface looks smooth and glassy

But

Is the flow structure laminar or turbulent?

What two ways could you determine this?

Visualise the flow structure by dropping some red dye into the flow- does it trace out a line or is it mixed?

Flow structure in the flume at the start of the experiment

The structure of the flow can also be determined by calculating the flow Reynolds number. All numbers must be in SI units (i.e., kg, m, s)!

The Reynolds is defined by:

$Re = Uh\rho/\mu$

where:
Re = Reynolds number
U = mean flow velocity
ρ = density of water
μ = molecular viscosity of water
h = mean water depth

For open channel flow a value less than 500 indicates laminar flow, a value greater than 2,000 indicates turbulent flow and a value in between these figures indicates transitional flow.

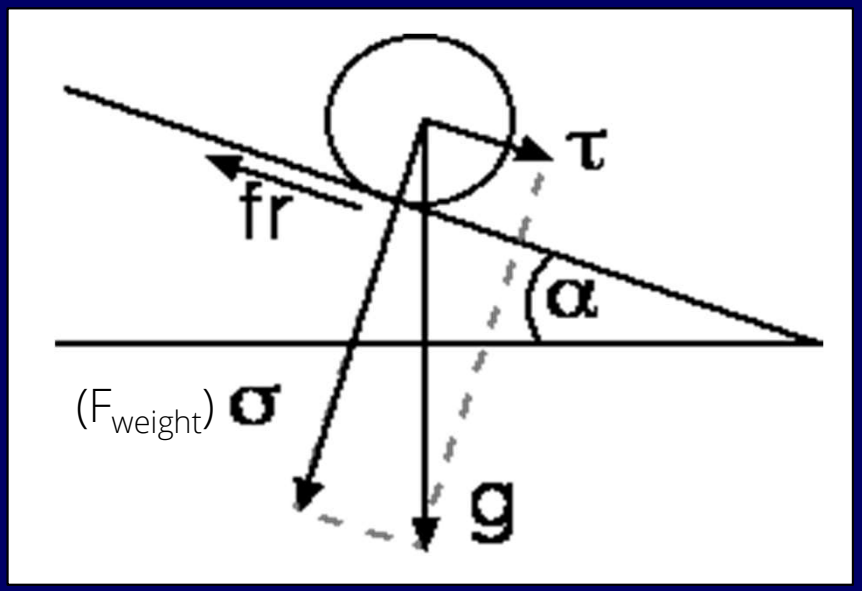
Answer: Re =

The results should demonstrate that all flows in the flume, and also in natural rivers, are fully turbulent even when the surface of the water is smooth and glassy

Conditions at the initiation of sediment transport

What was the critical shear stress (τ_{cr}) needed to get the grains to move?

Critical Angle (a) for grain movement



$$\tau_0 = \rho g h S$$

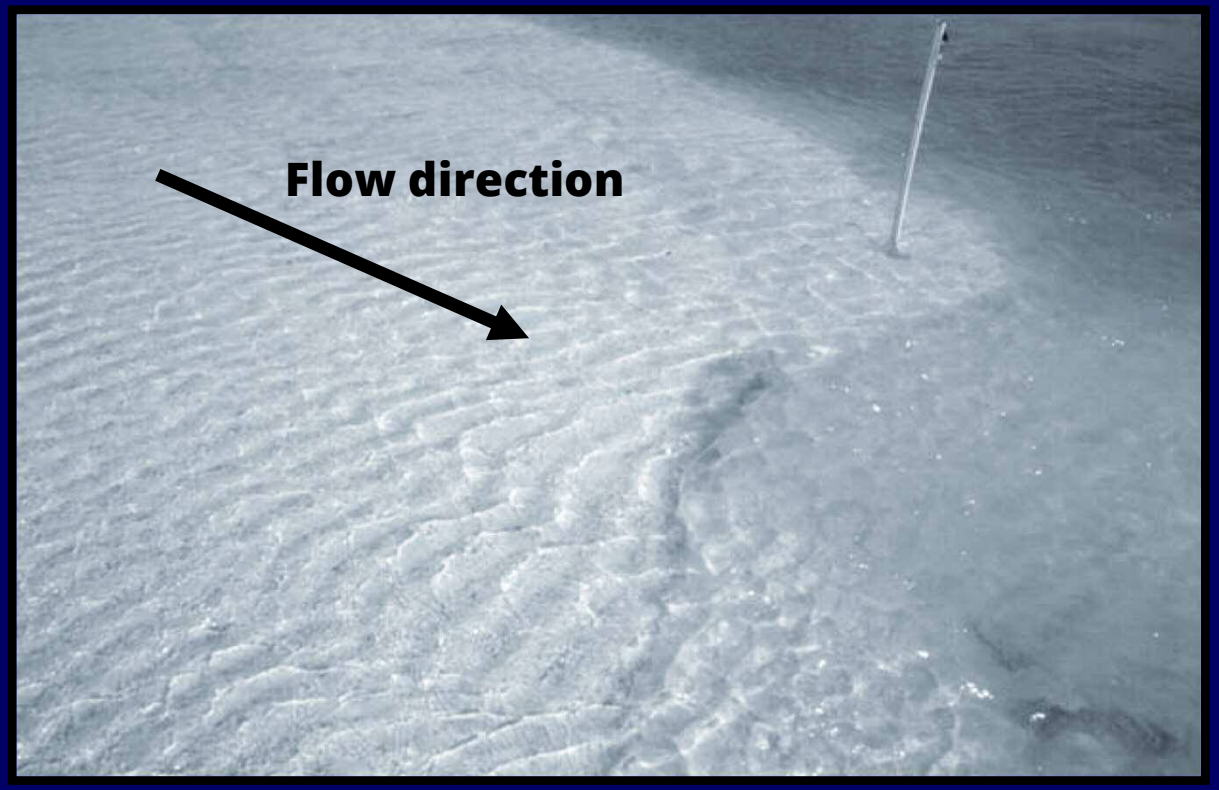
Critical angle (α) for motion is the slope where:

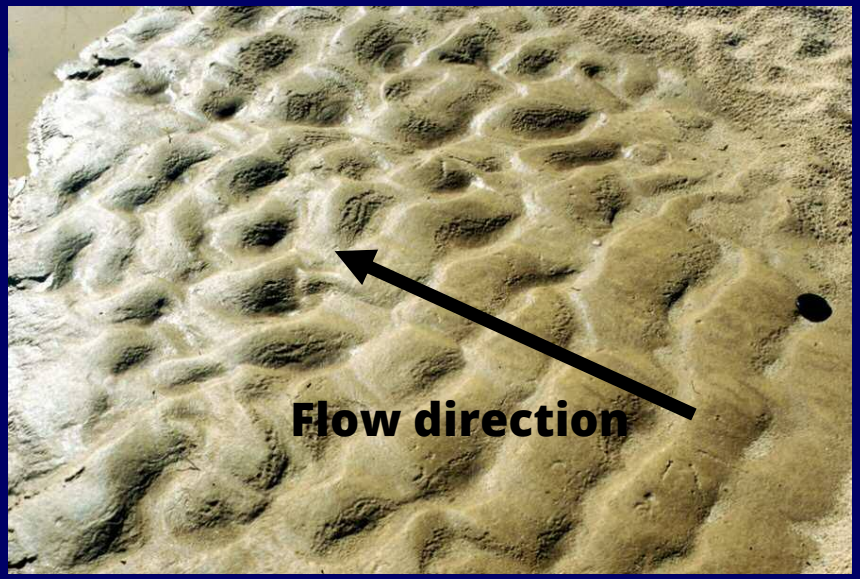
$$\tau = \sigma(fr)$$

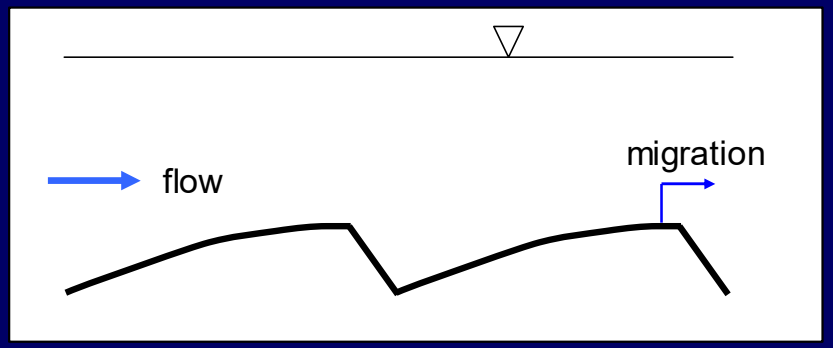
- an increase in slope leads to grain movement
- any decrease in slope will stop grain

In other words, at any slope (α) there is a critical shear stress (τ_{cr}) for movement of a grain

Current Ripples

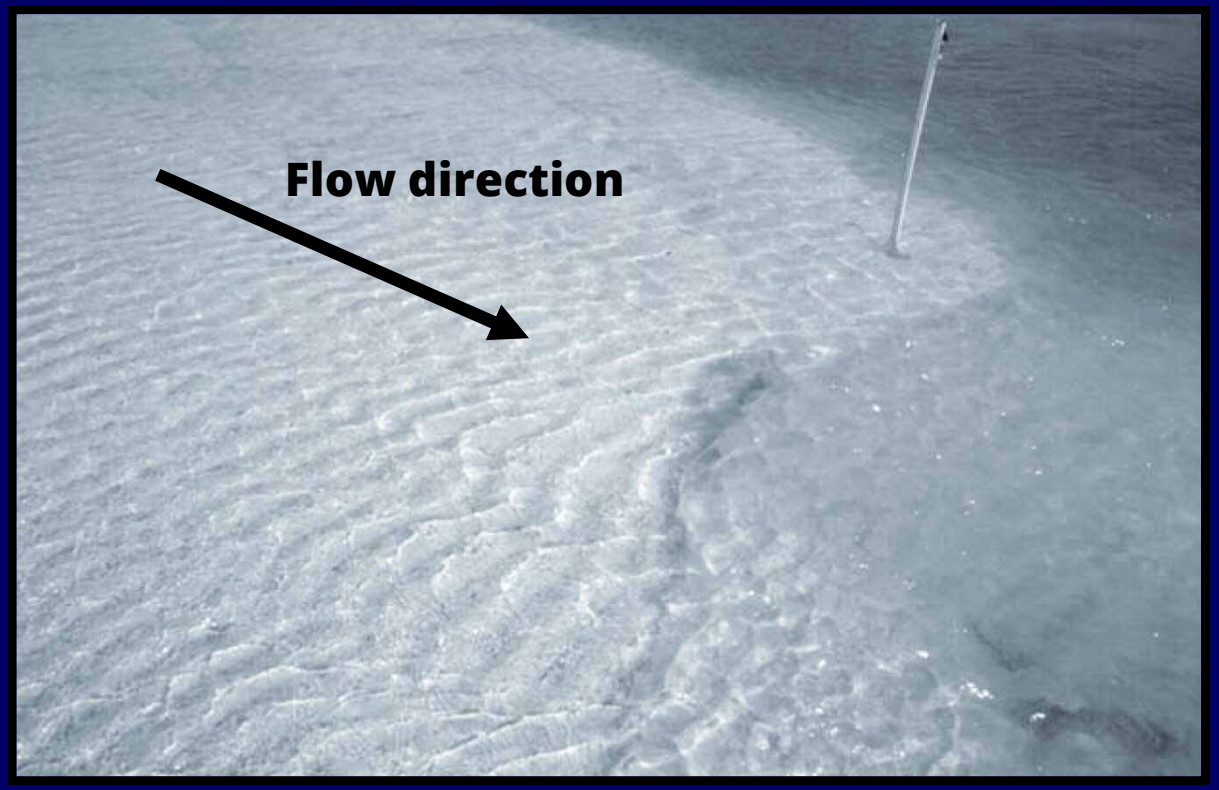


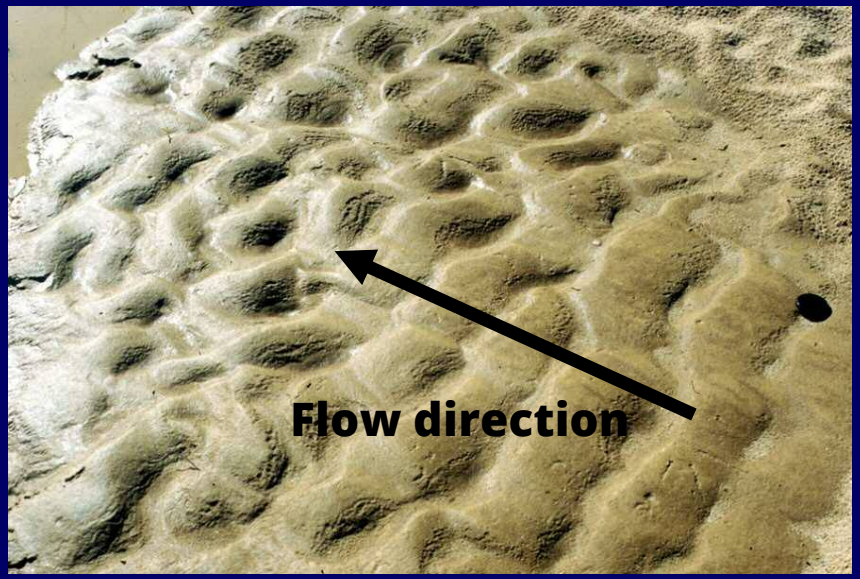


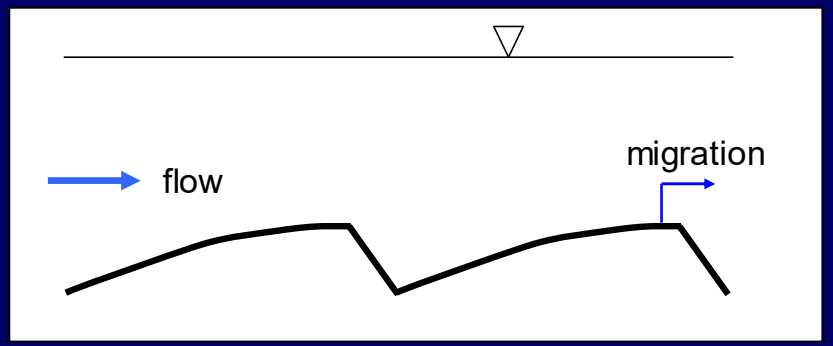


- Characteristic of very low transport rates (Form at threshold of bed load motion if hydrodynamically smooth)
- Mean grain size must be <0.5-0.7 mm (vf-m sand and some c sand)
- Migrate downstream, with sediment transport almost purely by bedload
- Small scale, asymmetric bedform (gentle stoss (upstream) and steep lee (downstream))
- Dimensions: heights 0.005-0.05 m, wavelengths 0.05-0.6 m

Current Ripples





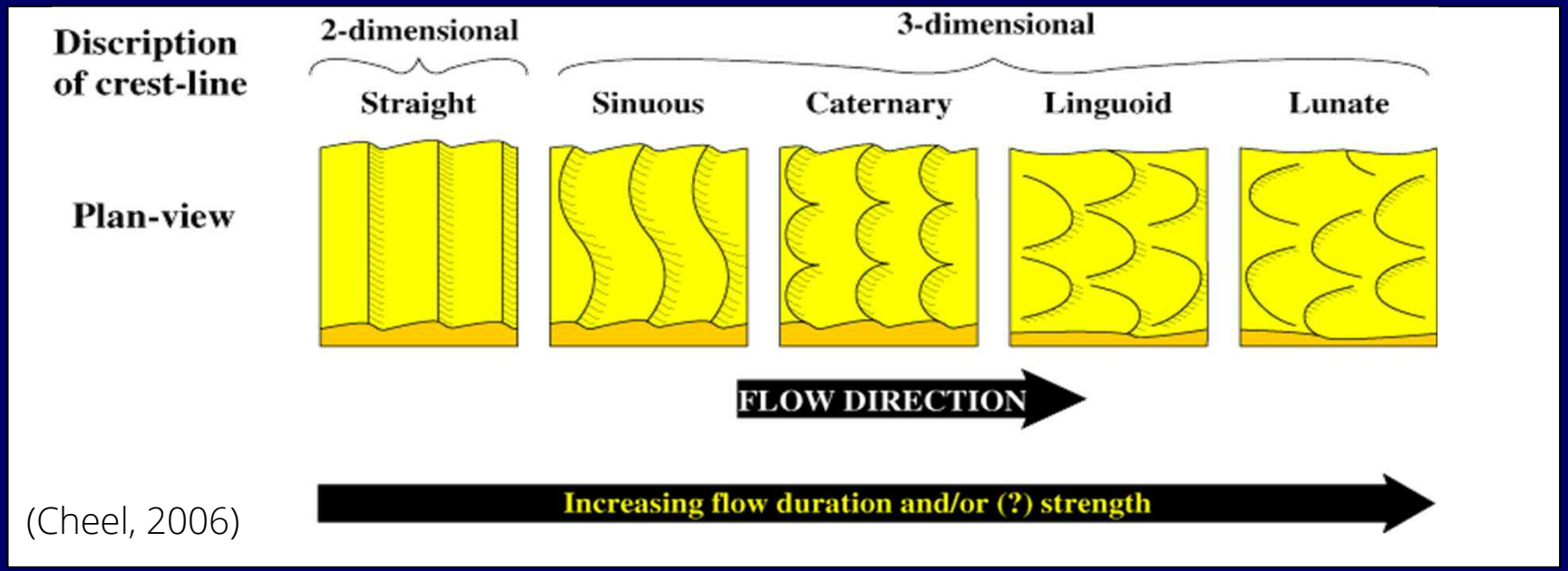


- Turbulent boundary must be hydraulically smooth for ripples to form
- Do not interact with water surface and no dependence on flow depth (No relationship between ripple height and either flow strength or water depth)
- Wavelength does vary with grain size (λ=1000d)
- Migrate downstream can be used to determine palaeocurrent direction

Ripple planform varies with flow strength and flow duration







(Cheel, 2006)

Current ripples geometry





- Experimental work suggests that fully developed ripples are always 3-D. The straight and sinuous forms are metastable always changing to linguoid with sufficient time.
- 2-D ripples indicate either:
    - flow was short-lived (ephemeral), or
    - ripples formed in presence of waves

Summary

We will conclude our experiment next week:

How do water flows interact with a bed made of sand?

- Describe the bedforms that develop in fine-coarse sands under unidirectional flows (dunes to antidunes)
- Why ripples and dunes migrate downstream

Important points to remember from this week

Lower plane bed only forms in coarse sand or coarser

Ripples form only in medium sand or finer
Ripple spacing: increases with grain size
No clear relationship between ripple size and either flow strength or water depth although planform may vary with flow strength and/or flow duration